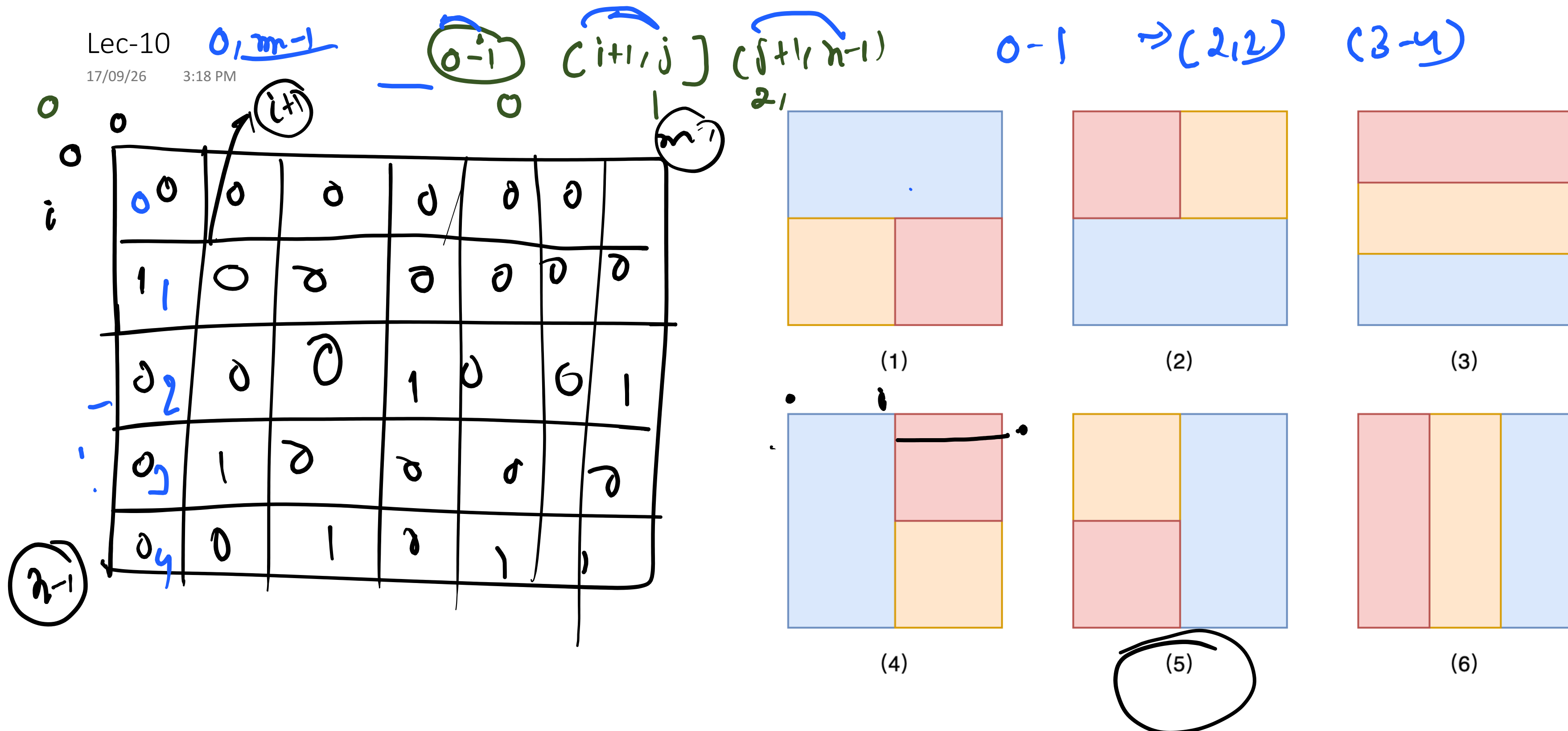
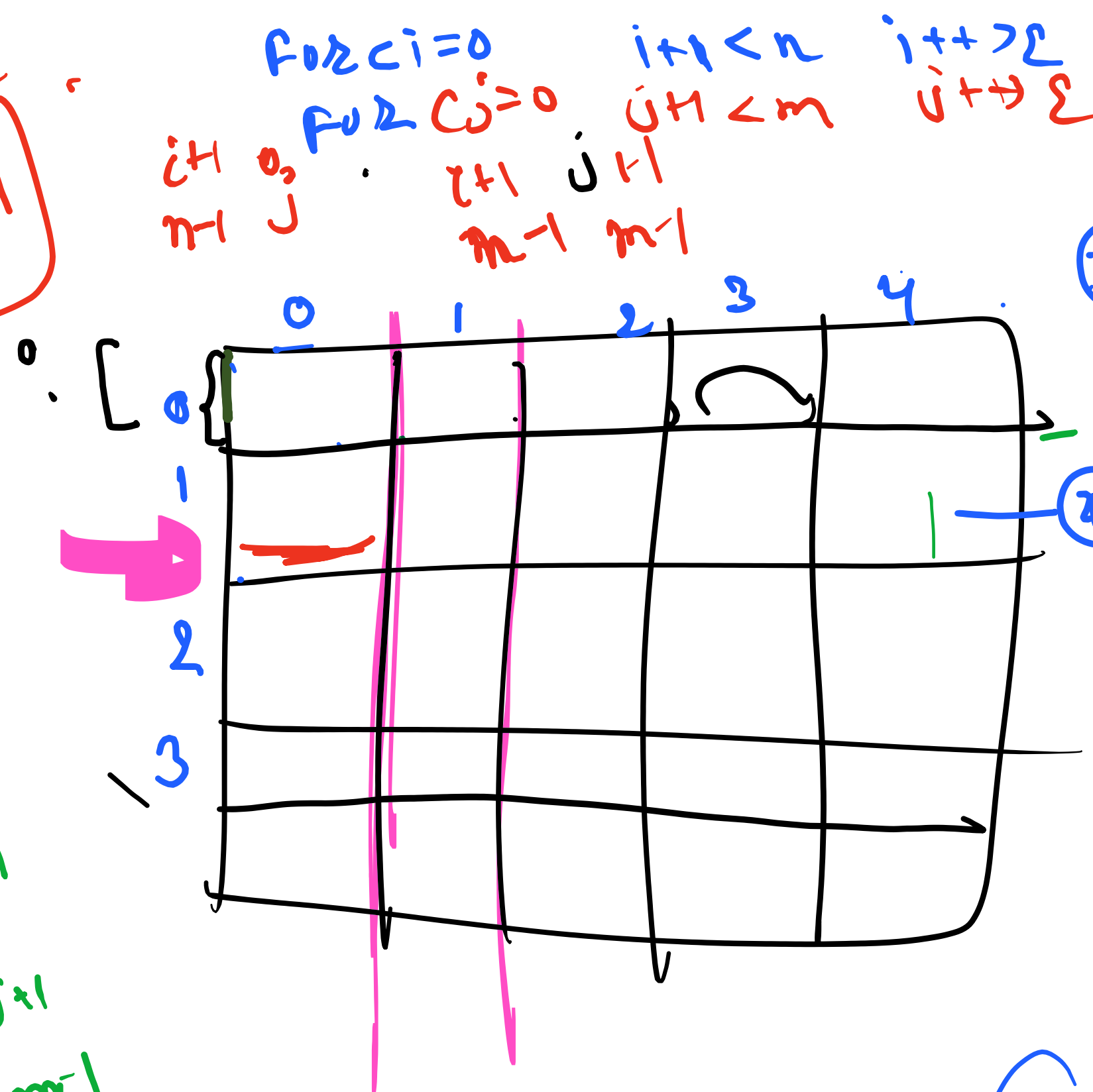
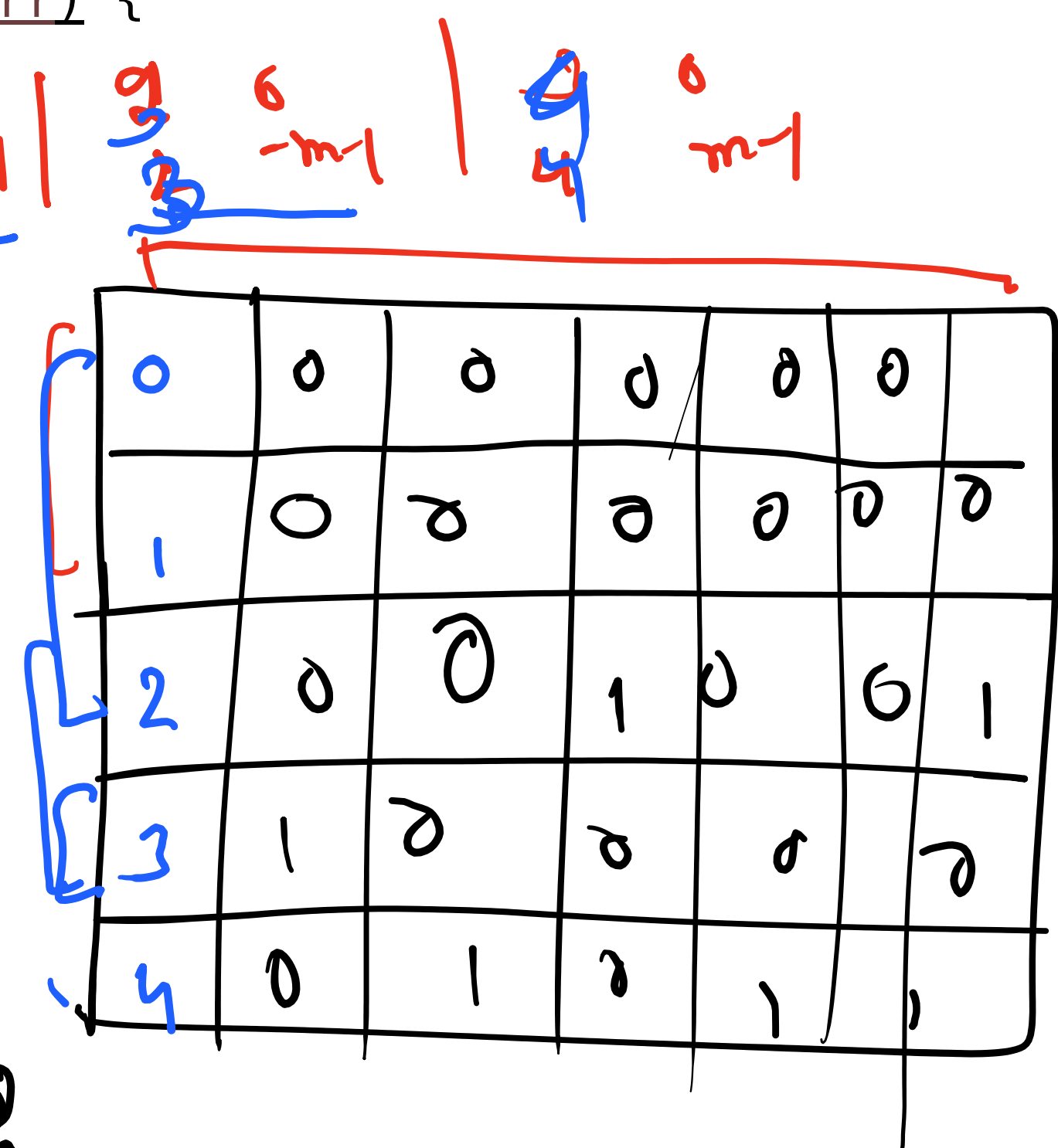




```
public static int Find the Minimum Area to Cover(int[][] arr) {
```

```
int n = arr.length;
int m = arr[0].length;
int ans=Integer.MAX_VALUE;
// case-1
for (int i = 0; i < n; i++) {
    for (int j = i+1; j+1<n; j++) {
        ans=Math.min(ans, res(arr, 0, 0, i, m-1)+
            res(arr, i+1, 0, j, m-1)+
            res(arr, j+1, 0, n-1, m-1));
```


$$\begin{array}{cc} 0 & 0 \\ i & j \\ \vdots & \vdots \\ n-1 & j \\ 0 & j+1 \\ n-1 & m+1 \end{array}$$
$$\begin{array}{c|c|c} 0-1 & 2-2 & 2-4 \\ 0 & & \\ 0- & \underline{1-2} & 3-4 \\ 0- & 1-3 & 4-4 \end{array}$$
$$\begin{array}{ll} \text{for } i = 0 & i+1 < n \quad i++ \quad \text{L} \\ p \leftarrow 0 & j+1 < m \quad j++ \quad \text{L} \end{array}$$